Name: Solutions

Further Maths
Core: Recursion & Financial Modelling
Mixed Practice SAC

Question 1

Jeremy is planning to save money for his "Schoolies" activity at the end of the year. He has an account with \$1500 in it already, and each month he will add another \$75 from his part-time job. The account pays 4.8% p.a. interest, compounded monthly.

a) What is the interest rate per month? (1 mark)

$\textit{Interest rate per month} = \frac{4.8}{12}\% = 0.4\%$

b) Using V_n to represent the balance of the account after n months, write a recurrence relation to model this investment. (2 marks)

Recurrence relation for the Annuity Investment: $V_0 = \textit{initial value}, \quad V_{n+1} = R.V_n + 75$

$$V_{n+1} = \left(1 + \frac{r}{100}\right).V_n + 75 = \left(1 + \frac{\frac{4.8}{12}}{100}\right).V_n + 75 = 1.004.Vn + 75$$

Recurrence relations: $V_0 = \$1500, \quad V_{n+1} = 1.004.V_n + 75$

c) Use your calculator to determine recursive the value of the investment after Jeremy has made six payments. (2 marks)

Month number	Calculation details	Value of account
0		1500.00
1	1500.00 × 1.004 + 75	1581.00
2	1581.00 × 1.004 + 75	(1662.324…) = 1662.32
3	1662.324… × 1.004 + 75	(1743.9732…) = 1743.97
4	1743.9732… × 1.004 + 75	(1825.9491…) =1825.95
5	1825.9491… × 1.004 + 75	(1908.2529…) = 1908.25
6	1908.2529… × 1.004 + 75	(1990.8859…) = 1990.89

The value of the investment after six payments is \$1990.89.

d) Jeremy will close the account after twelve (12) months. How much money, correct to the nearest cent, will he have for his "Schoolies" activities? (1 mark)

N	I(%)	PV	Pmt	FV	PpY	CpY
12	4.8	– 1500	– 75	?	12	12

*Solve for **FV** = 2493.671…*
He will have \$2493.67 to spend.

Question 2

Jeremy's Aunt Alice is about to retire and is wondering what to do with her superannuation money

One option is to invest her $625 000 in an annuity paying 4.5% p.a. compounding monthly.
a) How much would she receive per month if the annuity is for 20 years? (1 mark)

N	I(%)	PV	Pmt	FV	PpY	CpY
20 × 12 = 240	*4.5*	*– 625 000*	*?*	*0*	*12*	*12*

Please note that Alice is about to retire so the investment is referred to is the annuities where it would pay her a regular monthly income, which eventually will run out. Do not confuse this with an annuity investment, as it is mainly applied to the "saving" stage, that is saving for your superannuation.
*Solve for **Pmt** = 3954.058…*
She will receive $3954.06 per month.

b) If she decided that she wanted to receive $4000 per month, how many fewer payments would she receive from the annuity? (2 marks)

N	I(%)	PV	Pmt	FV	PpY	CpY
	4.5	*– 625 000*	*4000*	*0*	*12*	*12*

Solve for N = 235.570… ≈ 235 months. The annuity will last for 235 months with payment maintaining at $4 000 with $2276.58 left over for the last payment (236), which is 240 – 235 = 5 months fewer than in (a).

c) Another annuity company claims that their annuity product would give her $4250 per month for twenty years for the $62500 investment. What interest rate, correct to two decimal places, would give this return?

N	I(%)	PV	Pmt	FV	PpY	CpY
240	*?*	*– 625 000*	*4250*	*0*	*12*	*12*

*Solve for **I%** = 5.3599…*
The interest rate applying in this case is 5.36 % per annum

Alicia believes that either twenty-year annuity discussed above will not last long enough for her, so she looked at investing in a perpetuity.

d) What is the difference between an annuity and a perpetuity? (2 marks)
A perpetuity is an annuity that provides regular payments that continue forever whereas an annuity payments only last for a period of time and will eventually stop.

e) If the interest offered is 5.2 % p.a., compounding monthly, how much will Alicia receive per month, correct to the nearest dollar? (1 mark)

$$D = \frac{r}{100} \times V_0$$

$$\textit{Interest (payment)} = \left(\frac{\frac{5.2}{12}}{100}\right) 625\,000 = \$2708.333\ldots \approx \$2708$$

f) If Alicia wanted to receive at least \$3000 per month from this perpetuity, how much would she need to invest, correct to the nearest thousand dollars? (2 marks)

$$\text{Interest (payment)} = \left(\frac{\frac{rate}{12}}{100}\right) \times principal = \left(\frac{\frac{r}{12}}{100}\right).V_0$$

$$\text{Interest (payment)} = Solve\left(3\,000 = \left(\frac{\frac{5.2}{12}}{100}\right).V, V\right)$$

principal (p) ≈ \$692 307.692…

She would have to invest \$693 000 and will receive \$3000 per month!
(Investing \$692 000 would earn less than \$3000 per month – \$2998.67!)

Question 3

Jayde is a potter and has purchased a pottery kiln for \$6500. The kiln can be depreciated using the reducing balance method at the interest rate 17.5% per year.

a) Using V_n to represent the value of the kiln after *n* years, write a recurrence relation that models this depreciation situation. (2 marks)

$$\text{Recurrence relation: } V_{n+1} = \left(1 - \frac{r}{100}\right).V_n = \left(1 - \frac{17.5}{100}\right).V_n$$

$$V_0 = \$6500, \quad V_{n+1} = 0.825.V_n$$

b) Use your calculator to determine recursively the value of the kiln, each year, for the first five years, and fill in the values in the table below. (2 marks)

Year number	Calculation details	Value of account
0		6500.00
1	6500.00 × 0.825	5362.50
2	5362.50 × 0.825	(4424.0625…) = 4424.06
3	4424.0625… × 0.825	(3649.8515…) = 3649.85
4	3649.8515… × 0.825	(3011.1275…) = 3011.13
5	3011.1275… × 0.825	(2484.1802…) = 2484.18

c) If Jayde will trade in her kiln for a new one when this one has a value less than \$2500, explain why this will occur after five years. (1 mark)

At the end of the fifth year, the value of the kiln will be \$2484.18, which is less than the trigger value for replacement.

Jayde has been advised to look at using the prime cost or flat rate depreciation method using a value of 15% of the purchase price.

d) By what amount, correct to the nearest dollar, will the value of the kiln be depreciated each year? (1 mark)

$$D = \frac{r}{100} \times V_0$$

$$15\% \text{ of } \$6500 = \left(\frac{15}{100}\right).6500 = \$975$$

e) Using V_n to represent the value of the kiln after *n* years, write a recurrence relation that models this depreciation situation. (2 marks)

$$V_0 = \$6500, \quad V_{n+1} = V_n - 975$$

f) How many years will it take for the value of the kiln to depreciate to $2600.

Using $V_n = V_0 - nD$, *where* $V_n = \$2600$, $V_0 = \$6500$, $D = 975$

$2600 = 6500 - 975n$

$975n = 3900$

$n = 4.0$

Alternatively, using the solve function in the CAS

Solve $(2600 = 6500 - 975n, n)$

$n = 4.0$

It will take four (4) years to depreciate to a value of $2600.

One of Jayde's pottery colleagues depreciates her kiln using the unit cost method at the rate of $5.70 per kiln load, where the average number of loads per year is 175.

g) Using V_n to represent the value of the kiln after *n* loads, write a recurrence relation that models this depreciation situation for Jayde. (2 marks)

$V_0 = \$6500 \quad V_{n+1} = V_n - 5.70$

h) How long will it take in years, rounded to the nearest whole number, for the value of Jayde's kiln to depreciate to $2500 if it is depreciated using this method? (2 marks)

Find the number of loads that would depreciate the kiln to $2500, and then the number of years.

Using the general formula:

$V_n = V_0 - nD$ *where* $V_n = \$2500$, $V_0 = \$6500$, $D = 5.70$

$2500 = 6500 - 5.70n$

$5.70n = 4000$

$n = 701.754\ldots \approx 702$ loads

Alternatively, using the solve function in the CAS

Solve $(2500 = 6500 - 5.70n, n)$

$n = 701.754\ldots \approx 702$ loads

Number of years $= 702$ *loads* $\div 175$ *loads per year* $= 4.011\ldots \approx 4$ *years*

It will take four (4) years for the kiln to depreciate its value to $2500.

Question 4

Sam has recently bought a home. He borrowed $350 000 from a bank which is charging him 6.79% p.a., compounding monthly, to be paid over twenty years?

a) What is the effective interest rate, correct to three significant figures, that Sam will be paying on his loan at 6.78% p.a. compounding monthly? (1 mark)

$$r_{eff} = \left[\left(1 + \frac{r}{100 \times n}\right)^n - 1\right] \times 100 = \left[\left(1 + \frac{6.78}{100 \times 12}\right)^{12} - 1\right] \times 100 = 6.994707 \ldots \approx 6.99\%$$

$r_{eff} = eff(r, n) = eff(6.78, 12) = 6.994707 \ldots \approx 6.99\%$ - *using the in-built CAS function*

Where r is the nominal interest rate per year and n is the number of compounding periods per year

b) Use the Finance Solver function on your calculator to determine Sam's monthly payment, correct to three significant figures. Write the values used in the table below. (1 mark)

N	I(%)	PV	Pmt	FV	PpY	CpY
20 × 12 = 240	*6.78*	*350 000*	*?*	*0*	*12*	*12*

*Solve for **Pmt** = –2667.520…*

The monthly payment is calculated to be $2670.

c) What is the monthly interest rate that Sam will be charged, written as a decimal correct to three decimals places? (1 mark)

Monthly Interest rate: $= \frac{6.78\,\%}{12} = 0.565\%$

d) Using V_n to represent the balance of the loan after *n* months, write down a recurrence relation to model this loan situation. (2 marks)

Recurrence relations: $V_0 = initial\ value, \quad V_{n+1} = R.V_n - 2670$

$$V_{n+1} = \left(1 + \frac{r}{100}\right).V_n - 2670$$

$$V_{n+1} = \left(1 + \frac{\frac{6.78}{12}}{100}\right).V_n$$

$$V_{n+1} = 1.00565.Vn - 2670$$

Recurrence relations: V_0 = *$350 000,* V_{n+1} = *1.00565.*V_n *− 2670*

Part of the amortisation table for Sam's loan is shown below.

Payment number	Payment made	Interest paid	Principal reduction	Balance of loan
0	0.00	0.00	0.00	350 000.00
1	2670.00	(i) *1977.50*	692.50	349 307.50
2	2670.00	1973.59	(ii) *696.41*	348 611.09
3	2670.00	1969.65	700.35	(iii) *347 910.74*
4	2670.00	1965.70	704.30	347 206.44
5	2670.00	1961.72	708.28	346 498.16
6	2670.00	1957.71	712.29	345 785.87

e) Calculate to the nearest cent:
i) the interest that will be paid with the first payment? (1 mark)

$Interest = \left(\frac{\frac{6.78}{12}}{100}\right) \times 350\,000.00 = \$1977.50$

ii) the amount that is reduced from the principal with the second payment? (1 mark)

Principal reduction = payment – interest = $2670.00 – $1973.59 = $696.41

Principal reduction = previous balance – current balance = $349307.50 – $348611.09 = $696.41

iii) the balance of the loan after the third payment is made? (1 mark)

Balance of loan = previous balance – principal reduction = $348 611.09 – $700.35 = $347910.74

Balance of loan = previous balance + interest – payment = $348611.09 + $1969.65 – $2670 = $347910.74

f) Calculate the total amount of money paid by Sam for his first six payments? (1 mark)

Total paid = 6 × $2670 = $16020

g) How much has Sam actually paid off his loan after the sixth payment? (1 mark)

Amount paid off = Principal – Amount owed after 6 years (obtain from above table)

Amount paid off = $350 000 – $345785.87 = $4214.13

h) Calculate the percentage correct to the nearest whole number, of Sam's first six payments that have been used to pay the interest charges. (1 marks)

Interest paid within 6 payments = Total amount paid – Amount paid off = $16020 – $4214.13 = $11805.87

$Percentage\ (interest) = \frac{Interest\ paid\ in\ the\ 1st\ 6\ payments}{Total\ amount\ paid\ in\ the\ 6\ payments} \times 100 = \frac{11805.87}{16020} \times 100 = 73.69\% \approx 74\%$

i) The loan is to be repaid with a number of monthly payments of $2670.00 and a final payment that is to be adjusted so that the loan will be fully repaid after exactly 20 years of monthly payments. Calculate the amount of the final payments, correct to the nearest cent. (3 marks)

Show details of any Finance Solver calculations in the table provided below.

20 years = 20 × 12 = 240 months/payments

Need to calculate the amount still owing prior to the last payment, that is 239 payments

N	I(%)	PV	Pmt	FV	PpY	CpY
239	*6.78*	*350 000*	*– 2670*	*?*	*12*	*12*

Solve for FV = –1404.2404…

The balance owing after the second last payment is $1404.24.

Interest payable for the last payment $=\left(\frac{\frac{6.78}{12}}{100}\right) \times 1404.24 =$ *$7.93*

Final payment = $1404.24 + $7.93 = $1412.17